RAJ PATEL

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EDUCATION

University of the Pacific Aug 2022- Dec 2024
Master of Science in Computer Science

LDRP Institute of Technology and Research Aug 2017- May 2021
Bachelor of Engineering in Computer Engineering

SKILLS

Programming & Frameworks: Python, Java, C++, C
Python Libraries: LangChain, PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, XGBoost, NLTK, Keras, Matplotlib
ML & DL Architectures: NLP, CNN, Random Forest, Linear Regression, Logistic Regression, Clustering, SVM, Naïve Bayes
Data Visualization & Analytics: Power BI, Tableau, Excel
Cloud & Devops: AWS (EC2, S3, IAM, Sagemaker), Docker, CI/CD, Jenkins, Git
Databases & Big Data: MySQL, PostgreSQL, Hadoop, GitHub
CourseWork: Machine Learning, Deep Learning, Data Science, Cloud Computing, Probability & Statistics, Discrete mathematics, Data Structure and Algorithms, Object-Oriented programming, Operating Systems

EXPERIENCE

Menzies Mission Feb 2025- Present
Software Engineer, Machine Learning

- Deployed LangChain and Hugging Face Transformers NLP pipelines to build LLM-powered agents that summarize beneficiary feedback via REST APIs and JSON-structured field reports, reducing manual review time by 40%.
- Experimented with prompt engineering techniques using LLMs (including OpenAI GPT-compatible interfaces via LangChain) to automate orchestration of feedback summarization and AI-driven workflow routing.
- Improved NGO program efficiency by 30% by developing donor segmentation models (K-Means, Random Forest) in Python, Scikit-Learn, and TensorFlow, optimizing fundraising campaigns and resource allocation.
- Predicted donation patterns using time-series forecasting (ARIMA, LSTM) to support budget planning; tracked tasks and documented workflows using Jira and Confluence.

Power Master Industries Sep 2021-July2022
Data Analyst

- Improved production and reporting efficiency by around 25% by analyzing manufacturing and sales data using SQL (PostgreSQL) to identify process delays and resource gaps.
- Cleaned and standardized large datasets using Python (Pandas) and SQL, increasing data accuracy; developed interactive Power BI dashboards to visualize daily output, sales, and inventory status.

ShreeHari Technology June 2021-Aug2021
Machine Learning Intern

- Built Deep Neural Networks (DNN) to classify user behavior and predict churn, improving prediction accuracy by 15%.
- Processed and analyzed client business datasets using Python, Pandas, NumPy, and SQL, delivering insights through visual dashboards for decision-making.

ACADEMIC PROJECTS

Speech-to-Text Summarization | *Python, TensorFlow, AWS SageMaker*

- Reduced manual notetaking by 75% by building a speech-to-text summarization system converting lecture audio into concise notes with 90% accuracy.
- Developed an end-to-end NLP pipeline using Google Speech-to-Text API and transformer-based summarization models with JSON-formatted outputs; deployed on AWS SageMaker for real-time processing.

Honeypot Attack Classification & Forecasting | *Python, Scikit-Learn, ARIMA, LIME*

- Built Random Forest and XGBoost classification models on honeypot attack data; evaluated performance using ROC-AUC Curve and Confusion Matrix to identify the best model.
- Applied ARIMA forecasting to predict future attack timing and integrated LIME (Explainable AI) to identify key features driving predictions.

CERTIFICATIONS

- AWS Certified Cloud Practitioner | Link
- Machine Learning Specialization | Link